

Midterm Exam - Econometrics I 2025

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General Instructions:

- Write legibly and organize your answers clearly. We will not grade what we cannot read;
- Do not try to deceive us. If you know something, say it. If you don't, say you don't;
- If you feel you need extra assumptions, state them clearly;
- This exam has 12 points in total. But please note that the maximum grade is 10;
- If you decide to not tackle a given question, state that clearly;
- You have **four hours** to complete the exam. There will be no extensions;

Problem 1 – (points: 2.5)

Consider the nonparametric regression model:

$$Y_i = m(X_i) + u_i, \quad i = 1, \dots, n$$

where $\mathbb{E}[u_i|X_i] = 0$ and $m(\cdot)$ is an unknown smooth function. The **local-linear estimator** of $(m(x_0); m'(x_0))$ at a point x_0 is defined as the solution to the weighted least squares problem:

$$\{m_{LL}(x_0), m'_{LL}(x_0)\} \equiv \min_{\alpha, \beta} \sum_{i=1}^n K_h(X_i - x_0) [Y_i - \alpha - \beta(X_i - x_0)]^2$$

where $K_h(u) = \frac{1}{h} K\left(\frac{u}{h}\right)$ is a second-order continuous kernel function with bandwidth $h > 0$. You have a random sample of size n : $\{(Y_1, X_1), \dots, (Y_n, X_n)\}$. We also assume that $X_i \in \mathbb{R}$.

- (a) **(0.5 points)** Show that the local-linear estimator can be written as a Generalized Least Squares (GLS) estimator. Specifically, show that there exists an $n \times 1$ vector V , an $n \times 2$ matrix Z , and an $n \times n$ matrix W such that:

$$\begin{pmatrix} m_{LL}(x_0) \\ m'_{LL}(x_0) \end{pmatrix} = (Z'W^{-1}Z)^{-1}Z'W^{-1}V$$

- (b) **(0.75 points)** What happens to W if, for a given n , we have that $h \rightarrow \infty$?
- (c) **(0.75 points)** When deriving the asymptotic theory for the local-constant estimator, we assumed the bandwidth was such that $h \rightarrow 0$ and $nh \rightarrow \infty$ as $n \rightarrow \infty$. Explain intuitively why each of these two conditions were necessary.

- (d) **(0.5 point)** Are such conditions also needed for the asymptotic theory of the local-linear estimator? What are the main differences between these two estimators? Briefly explain.

Problem 2 – (points: 2)

Consider the linear regression model:

$$Y_i = X_i' \beta + u_i, \quad i = 1, \dots, n$$

where $X_i \in \mathbb{R}^k$ and $\mathbb{E}[u_i|X_i] = 0$. You have a random sample $\{(Y_i, X_i)\}_{i=1}^n$, and the OLS estimator is $\hat{\beta}_n = (X'X)^{-1}X'Y$ where X is the $n \times k$ design matrix and Y is the $n \times 1$ vector of outcomes. You are interested in making inference about β .

- (a) **(1 point)** Describe the non-parametric bootstrap algorithm we can use to approximate the distribution of $R_n = \sqrt{n}(\hat{\beta}_n - \beta)$. Be specific about:
- What do you resample (what are the "units" being resampled)?;
 - Should you use replacement or not?
 - At each run of the bootstrap, how you compute the bootstrap OLS estimator?
- (b) **(0.5 points)** Suppose you are interested in testing $H_0 : \beta_j = 0$ for some coefficient $j \in \{1, \dots, k\}$. You ran the bootstrap with $B = 10,000$ replications and obtained bootstrap t-statistics $\{t_n^{(b)}\}_{b=1}^{10,000}$ where:

$$t_n^{(b)} = \frac{\hat{\beta}_{j,n}^{(b)} - \hat{\beta}_{j,n}}{\text{SE}^{(b)}(\hat{\beta}_{j,n}^{(b)})}$$

Explain how you would use these bootstrap statistics to conduct a two-sided test of H_0 at significance level α using this t -statistic.

- (c) **(0.5 points)** Suppose you ran the pairs bootstrap with $B = 10,000$ replications and obtained bootstrap statistics $\{R_n^{(b)}\}_{b=1}^{10,000}$ where $R_n^{(b)} = \sqrt{n}(\hat{\beta}_n^{(b)} - \hat{\beta}_n)$. Let $c_{\alpha/2,j}$ and $c_{1-\alpha/2,j}$ denote the $\alpha/2$ and $1 - \alpha/2$ quantiles of the j -th component of these bootstrap statistics. Construct a $(1 - \alpha)$ confidence interval for β_j using these quantiles.

Problem 3 – (points: 2)

Answer the questions below about ARMA processes. Please notice that the items are not related to each other.

- (a) **(1 point)** Suppose that $Y_t = X_t + e_t$ and $X_t = aX_{t-1} + u_t$, where e_t and u_t are mutually independent i.i.d. random variables with finite variance and zero mean, and $|a| < 1$. Show that Y_t has the same autocovariance function as a certain ARMA(1,1) process. What are the AR and MA coefficients of this process in terms of a and the variances of e_t and u_t ?

Hint: Use the MA(∞) representation of X_t .

- (b) **(1 point)** Consider the AR(3) process:

$$Y_t = -0.22 + 0.3Y_{t-1} + 0.5Y_{t-2} + 0.2Y_{t-3} + \varepsilon_t$$

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

- (i) Is this process stationary? If so, compute its entire autocovariance function. If not, explain why.
- (ii) Determine the dynamics of $\Delta Y_t \equiv Y_t - Y_{t-1}$. What kind of process does it follow? Is it stationary?

Hint: Use the Fundamental Theorem of Algebra.

Problem 4 – (points: 3)

Consider a stationary process Y_t such that

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots$$

where $\varepsilon_t \sim i.i.d. N(0, \sigma^2)$, μ is known, $\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots$ is invertible (in the sense that $\theta(L)^{-1}$ is well-defined), and $\sum_{i=1}^{\infty} |\theta_i| < \infty$. You observe the data $\{Y_1, \dots, Y_T\}$. We will also assume that $\varepsilon_t = 0$ for $t \leq 0$ and that $Y_0 = \mu$.

- (a) **(0.25 points)** Let $\mathcal{F}_t$ be the information set generated by $\{Y_s : s \leq t\}$ and let $\mathcal{E}_t$ be the information set generated by $\{\varepsilon_s : s \leq t\}$. Argue that, in this case, $\mathcal{F}_t = \mathcal{E}_t$ for all t , i.e., they carry the same information.
- (b) **(0.75 points)** Suppose you want to create a forecast for Y_{t+h} with information up to time t , for some fixed $h > 1$. You decide to forecast this series using its conditional expectation $\hat{Y}_{t+h|t} \equiv \mathbb{E}[Y_{t+h} | \mathcal{F}_t]$. Define the forecast error as $e_{t+h|t} = Y_{t+h} - \hat{Y}_{t+h|t}$. Write an expression for this forecast error in terms of the shocks $\{\varepsilon_t\}$. Is $e_{t+h|t}$ a stationary process? Does the autocovariance structure of $e_{t+h|t}$ depend on the parameters θ_i ?
- (c) **(0.75 points)** Now consider a regression framework where you use an exogenous variable X_t to forecast Y_{t+h} :

$$Y_{t+h} = \beta_0 + \beta_1 X_t + u_{t+h}$$

where $\mathbb{E}[u_{t+h}|X_t] = 0$. You estimate this regression using OLS with data from $t = 1, \dots, T - h$. Show that u_{t+h} might exhibit serial correlation. What is the nature of this serial correlation (i.e., what is the correlation structure)?

- (d) **(0.75 points)** Given your answer to part (b), explain why standard OLS standard errors would be invalid in this regression. What type of standard error estimator should you use instead? Be specific about what parameters (if any) need to be chosen when implementing this estimator.

Problem 5 – (points: 2.5)

Consider the linear regression model:

$$Y_t = X_t' \beta + u_t, \quad t = 1, \dots, T$$

where X_t is a $k \times 1$ vector of regressors and β is a $k \times 1$ vector of parameters. Assume that $\mathbb{E}[u_t X_t] = 0$ (the moment condition), but you are concerned that the errors might be heteroskedastic and/or serially correlated.

- (a) **(0.5 points)** Define the sample moment function $g_T(\beta) = \frac{1}{T} \sum_{t=1}^T X_t(Y_t - X_t' \beta)$. Write down the GMM objective function that you would minimize to obtain the GMM estimator $\hat{\beta}_{GMM}$ using a weighting matrix W_T . What is the first-order condition for this minimization problem?
- (b) **(0.5 points)** Suppose initially that the errors are i.i.d. with $\mathbb{E}[u_t^2|X_t] = \sigma^2$ (homoskedastic). Derive the optimal weighting matrix W^* for the GMM estimator. Show that with this optimal weighting matrix, the GMM estimator coincides with the OLS estimator.
- (c) **(0.75 points)** Now suppose the errors are heteroskedastic: $\mathbb{E}[u_t^2|X_t] = \sigma_t^2$ (which may vary with t). What is the optimal weighting matrix in this case? How would you estimate this weighting matrix in practice?
- (d) **(0.75 points)** Finally, suppose the errors are both heteroskedastic and serially correlated (but the correlation decays as the lag increases). What is the form of the optimal weighting matrix now? Describe how you would estimate this matrix using a HAC estimator. What choice(s) do you need to make when implementing this estimator?